October 26, 2026

Editor
Journal of Applied Physics

Dear Editor,

I submit the manuscript "Special optimal thermal cells: D-optimal thermal displacements for finite-temperature effective-harmonic phonons" for consideration as a regular article in the Journal of Applied Physics.

Finite-temperature phonons underlie applied predictions of phase stability, vibrational thermodynamics, and thermal transport, and the number of first-principles force calculations is the main cost of obtaining them. The manuscript asks whether designed, rather than randomly sampled, supercell displacements can reduce that cost, and answers it with fully matched controls. For eleven crystals of five structure classes (fcc Al, Ag, Au, Cu; bcc Mo, W; rocksalt NaCl, MgO, PbTe; diamond Si; hcp Mg) the designs are compared with converged large-cell references, with effective force constants fitted to canonical ensembles in the same cell, and, at identical numbers of force calls, cells, electronic settings, and fits, with the established alternatives: canonical sampling with TDEP-type least-squares fitting, its iterated form sTDEP, TDEP fitted to *ab initio* molecular dynamics (Al), SSCHA (with DFT forces for Al and on DFT-fitted anharmonic model potentials), special displacements, and thermal rattle with hiPhive regression.

The principal results are:

- a D-optimal antithetic design (SOTC-D) with an estimator theory: the effective force constants are the Gaussian-averaged Hessian, SOTC-D estimates them with an $O(1/M)$ bias at fourth order, and inversion pairs cancel cubic forces exactly;
- with the same trial, SOTC-D has the lowest dispersion error of five designs in 8 of 11 crystals at two and 9 of 11 at eight force calls; its bootstrapped form beats one-shot canonical sampling in all eleven crystals at every budget from 4 to 16 calls (supercell basis) and two-iteration sTDEP with DFT forces in all eleven at eight calls, and on anharmonic model potentials of five crystals its fully self-consistent form reaches a given accuracy with a factor of 2–16 fewer force calls than SSCHA in 13 of 14 cases;
- with DFT forces in Al and Si, scored against a converged self-consistent reference, SOTC-D-SC is compared with iterated sTDEP, self-consistent special displacements, SSCHA, and AIMD-TDEP: it is the most accurate at small budgets for strongly renormalized Al, whereas for weakly anharmonic Si pooled sTDEP and self-consistent special displacements are more accurate at 32 calls;
- regression method (nine least-squares, sparse, and Bayesian fits) matters little, whereas a crystal-symmetry basis reduces the parameter count of the cubic crystals by up to a factor of 30 at no loss of accuracy;
- a quantitative separation of sampling error from finite-temperature renormalization (up to 1.4 THz) and supercell truncation (up to 0.5 THz), room-temperature Debye–Waller factors within 6% of tabulated values (converged references for Al, Cu, and Au; SOTC-D models within 2–11%), and non-analytic LO–TO corrections for the polar crystals;

- on a volume grid, a reproducible thermal expansion of Al within 3% of experiment at 300 K and, with a cubic free-energy fit, 7–8% below it at 400 K, with a statement of where the small budgets of the design stop being sufficient (the free-energy curvature at higher temperatures), and for PbTe, whose small-cell harmonic FC2 is unstable, a stable self-consistent result about 20% below experiment at 300 K.

All inputs, outputs, and analysis scripts are openly available.

The manuscript is not under consideration elsewhere. Suitable reviewers include experts in first-principles anharmonic lattice dynamics and force-constant fitting methods.

Sincerely,

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